MULTIMEDIA UNIVERSITY

Faculty of Information Science and Technology

Intelligent QA System for Dengue-related Information

Lysa Liew Wan Jin [1], Kai Qing Lim [1], Xin Ping Ng [1], Kai Chun Teh [1]

[1] *Faculty of Information Science and Technology, Multimedia University, Melaka, 75450 Malaysia*

Abstract

This project intends to build a smart question-answering (QA) system that can provide reliable and multilingual dengue related information. The system reads a manually curated sample of 149 questions and answers in the Malay language and through NLP-based techniques translates it into English. RDFLib builds a structured knowledge graph, a graphic model of semantic correlations between categories, questions, answers, and sources. It is a knowledge base stored as Turtle and allows more advanced querying based upon SPARQL. To visualize and make sense of the data, the graph is plotted based on NetworkX and Matplotlib. It provides the possibility of flexible user input that is multilingual and includes automatic translation and categorisation. All in all, the project presents good use of knowledge representation and NLP in providing wide access to the population of dengue health information via semantically conscious QA interface.

1 Introduction

Smart question answering (QA) systems have been found to be an effective way of providing precise and accessible information to users in most fields and most importantly in the healthcare sector. As the world attention turns towards the health-related problems, dengue fever being one of the most threatening conditions. It becomes crucial to have some artifacts that can help not only with ease of using, but also with synonymity of language and credibility of medical information. In this project, the idea that is going to be developed is the smart QA system that is strictly adapted to deliver dengue-related information in both Malay and English languages.

The essence of the system is to allow users to pose an inquiry in both languages and get an answer back that is precise and appropriate due to the usage of manually curated datasets. To facilitate effective communication and understanding, the system employs the application of natural language processing techniques in translation, classification, and semantic matching. The number of prepared questions and answers makes 149 in the bilingual dataset, and they are clean, and ready to be used in downstream processing.

To express the information in Graphical and structured form, in a machine-readable format, a knowledge graph is created by using RDF (Resource Description Framework). This knowledge graph stores questions, answers, categories, and source URLs in a semantic fashion. It is SPARQL enabled and, as such, the data can be graphically plotted with the help of graph plotting tools, which makes it easier to determine relations between the concepts involved.

Moreover, the system incorporates Transformer-based model in multilingual semantic retrieval, which means that it compares user queries to the corresponding answers even with a different style of words. The system can be accessed in any web browser and does not require further installations since the user interface is run via Gradio and hosted on Hugging Face Spaces.

On the whole, this smart QA system shows that a combination of multilingual processing, structured knowledge representation, and problem-free deployment can boost the availability of essential dengue-related health information to the general public. It can be used as a basis for further advances in the field of domain-specific, multilingual health-related information systems.

2 Related Works

Deepa et al. (2024) [1] demonstrated that this research is done to improve the performance of recommendations in medical question-answering system through the fine-tuning of GPT-3.5 Turbo-0125 model on the MedQuAD dataset. The data is collected from NIH websites which consists of high-quality question answer pairs that ensures both diversity and reliability of medical content. As for data preprocessing, each question is concatenated with its answer to create proper text sequences and standardisation of input length to maintain uniformity and optimise the training process. This is to ensure that the model can generate accurate relevant answers. Byte Pair Encoding (BPE) tokenization is used to manage vocabulary size and post tokenization is used for conversion of text into numerical formats for model training. The model is initialised using the OpenAI API and fine-tuning process is done along with hyperparameter optimisation to maximise learning efficiency. This process improves a model's capability to generate outputs consistently, which is important for applications such as medical diagnostics and any therapy advice. It supports bigger input sizes of up to 4k tokens, making it ideal for complicated clinical queries. The performance of this system is assessed using BLEU, and ROUGE. This system shows excellent improvements in generating recommendations. However, the future work is to explore fine tuning process on other advanced language models.

Alruwaithi & Alhumoud (2025) [2] researched that the objective of SHIFA is to improve the existing Arabic healthcare question-answering system by using SBERT and hybrid deep learning models. The dataset is collected from Altibbi platform which has many latest and simplified medical information including questions and answers in Arabic. As for the data preprocessing phase, systematic cleaning is applied such as removal of diacritics, elongation characters, special symbols, URLs, numbers, stop words and non-Arabic characters. The process of text normalisation and standardisation of punctuation is also done. The dataset is then split into training, testing and validation sets. In the development of model, several pre-trained language models are used to integrate with SBERT. Therefore, a hybrid model is developed such that SBERT is input into classifier models like LSTM, GRU and CNN to classify the relevance of answer. The optimisation of hyperparameters is done and the model of performance is evaluated using accuracy, f1-score, precision and recall. The strength of this system is it achieve high accuracy as compared to other models. The future work will be implementing this model in other tasks such as question similarity, sentiment analysis and other relevant task. The use of multilingual corpus can also be considered in the SHIFA model for open domain.

Sukhwal et al. (2024) [3] stated that this research is designing an automated question-answering (QA) system for disease-related queries. This is developed for users by combining large language models (LLMs) and knowledge graphs to answer questions appropriately. The data collection phase involves obtaining disease information such as overview, symptoms, risks and other relevant information from NIH websites. The data preprocessing step involves the detection of abbreviation using Scispacy library which is designed for medical text. Conference resolution is also implemented to reduce ambiguity in entity references and structuring the data as informative triples for knowledge graph creation. The knowledge graph is built using Python programming language to ensure that it is suitable for the development of model rather than using the existing knowledge graph. The process of fine-tuning a sentence transformer is done through the use of unsupervised technique known as GenQ which is used to generate question and answer pairs. Large Google T5 LLM is used to implement the GenQ in order to specifically trained the process of generating QA pairs from given text body. The workflow of combining semantic retrieval through the process of finding relevant documents through the fine-tuned transformer and production of multiple candidate answers are carried out. The performance of system is evaluated based on metrics like ROUGE-L and BERTScore. This system effectively combines KG and LLM to provide high quality answers. The limitation of this system is it does not have the ability to update with the latest advances as it uses past knowledge. The future work is adding and design user interface component to make it more accessible for users.

According to Q. Peng et al. (2024) [4], this research develops a medical question-answering system named MeDQA which answer medical-related questions based on multi-source data. The data collection phase involves the gathering of images and medical data such as physical record of certain patient during their medical visit. In data preprocessing, the patient records are anonymised and the diverse clinical data in text and image format are systematically organised in JSON format. GPT-4 is used to generate high quality QA pairs using the JSON data as input. Three domain experts are arranged to review the generated questions to maintain data quality as the size of data that needs to be filtered is too large. The data is then split into training and testing set which is then used to fine-tune LLaMA-7B. The duplication of generated questions is also removed so that the data has unique questions. The model deployed is a multi-agent framework which is named as Medical Multi-Agent (MMA) system. GPT-4V is used to implement interfaces for each agent as they are assigned in handling a specific task and achieve relevant diagnosis result. The evaluation is done by comparing the existing approaches such as transformer-based model with the developed MMA model. The strength of this model is its ability to assign different roles to solve several medical tasks which produce structured decision-making output. However, the agents need to learn more medical knowledge in order to solve complex medical tasks.

Alkhurayyif (2025) [5] developed an Arabic open-domain medical question answering system (OQAS) for health information. The proposed OQAS integrates Graph Convolutional Networks (GCN) and Transformers, GPT 2.0 to improve the performance. This system begins with data acquisition, which uses the MAQA dataset that contains 430,000 queries and answers in Modern Standard Arabic (MSA) from 20 medical specialists. The data preparation phase is important to ensure accurate entity and relationship extraction and representation in the following stages. Preprocessing

includes discretization, reducing orthographic ambiguity, tokenisation, and morphological analysis. Besides, the normalisation techniques like stemming and lemmatisation are used to improve the text processing speed and model responsibility by reducing the word form variations. For Entity-Relationship (ER) extraction, this QA system identifies several of medical entities such as diseases, medications, symptoms, procedures, and healthcare providers, as well as their relationships. To categorise content, inquiry, and text, the authors utilised the multinomial Naïve Bayes (MNB) classifier. The system combines a GCN for modelling local relations between medical entities and a fine-tuned GPT 2.0 Transformer for global language understanding, which has been optimised with Hyperband. Their embeddings are fused through element-wise addition that use both global and local context to enhance the answer generation. The performance is evaluated using standard metrics like accuracy, precision, recall, F1-score, AU-ROC, AU-PRC, MAP, and Kappa coefficient.

In another study, Das & Nirmala (2022) [6] suggested a healthcare question answering system that can answers questions from healthcare contexts accurately. The SQUAD V1 dataset is used for training after performing feature extraction. This dataset consists of 107,785 question-answer pairs from 536 Wikipedia articles. It was split into 80% for training set, 10% for development set, and the other 10% for testing set randomly. The answers were categorised into numerical and non-numerical sequences for processing with Stanford CoreNLP. The SQUAD data was preprocessed into model-ready features by tokenising examples using WorldPiece and configuring attributes like max_seq_length, max_query length, doc stride, and padding. Hugging Face's squad processor then converts these into structured features for model input. The pre-trained BioBERT model that trained on the PubMed and the PMC data, is available at Hugging Face. In addition, this pre-trained model uses BERT's architecture to process masked words and biomedical terms. It combines the questions and context into a single sequence with segment embeddings, which A for question and B for context. Contextual embeddings are then generated for both questions and context inputs to facilitate the question-answering tasks. For each question-context pair, the model can compute the best answer span based on start and end token scores. The two evaluation matrices used in this paper are F1 score and Exact Match (EM).

Vinod et al. (2021) [7] presented a low-resource, multilingual, and a preliminary first-point-of-contact medical assistant model. The proposed model combines BioBERT, LABSE, and GPT-2 for multilingual medical question answering and information retrieval. For data collection, a large QA dataset, which contains 325,000 unique pairs from Reddit, WebMD, and Question Doctors, is encoded via BioBERT for similarity-based retrieval. An EHR (Electronic Health Record) summariser takes the patient history as input and summarises any current medications or treatments. A text pre-processing pipeline that begins with the tokenisation and the creation of embeddings for medical terms is conducted. Special tokens are added to support language modelling and the vocabulary used covers a wide range of clinical and biomedical terms. Additionally, sentences are embedded and assigned a positive integer IDs to facilitate downstream retrieval and translation tasks. The sentence is then translated into English for the BioBERT model using the LABSE model. For model training, BioBERT initialises the BERT model with shared weights for question and answer embeddings, which is a method inspired by DocProduct. GPT-2 enriches the retrieved answers and is trained using cross-entropy loss based on embedding similarity to generate the answer. LABSE translates the GPT-2 outputs into the target language, while DistillBERT is employed to cluster EHR summaries to support diagnostic processes. After that, the model was evaluated on BioASQ datasets (4b, 5b, 6b) using strict accuracy metrics.

A healthcare information provision question and answer system using BERT-GPT2 hybrid model is proposed by Jeong et al. (2023) [8]. The source data is collected from Jisik-in, a widely used question and answer platform in Korea affiliated with the Korea Medical Association. The domain focus is on Orthopedics, which uses three searching keywords such as joint pain, arthritis, and edema Korean. From over 48,039 question-answer pairs collected, 5,000 pairs were randomly selected and answered by real doctors. Each data sample contains four fields, including title, question, answer, and keyword, with the title and question fields merged. Text cleaning was applied to remove non-verbal characters, special symbols, English text, HTML tags, and redundant greetings or self-introductions. To standardise Korean spelling and spacing, a 'Pi-Hanspel' and 'Kospacing' library is used. Besides, all gaps and inconsistencies in the text were resolved for better model learning. BERT is used as an encoder for intent classification and keyword extraction from input queries, while GPT-2 acts as a decoder to generate contextually relevant responses and produce answer sentences. Pre-trained BERT and GPT-2 were integrated and optimised in this study. For evaluation, the performance of model was measured using perplexity (PPL), loss and a qualitative assessment.

Nguyen et al. (2022) [9] presented SPBERTQA, a two-stage question answering model, which is specialised in processing medical texts to better the accuracy of answers, with semantic comprehension. The initial phase of the system will be to obtain the semantically specific excerpts of a medical corpus with the help of Sentence-BERT to carry out the comparison of a dense vector embedding of a user query and sentences in a document. The second step involves applying a well-trained BERT model to pick up the ultimate response in these passages. The two-tier architecture is especially useful in solving the difficulties of medical text, which include intricate phrasing, domain-specific words, and very sparse answer contexts. The authors checked their model on the BioASQ dataset and received higher results than BERT-based baselines, especially at answer retrieval precision and relevance. Besides, the modular structure of the system enables its adaptation to other medical QA tasks or incorporation of any new retrieval strategies. But the sentence-based retrieval granularity continues to be a limitation of SPBERTQA when a contextual cue to an answer is distributed between sentences or even documents. This may affect the work at higher levels of fine-grained or multi-sentence

arguments. Still, SPBERTQA can be discussed as a good contribution to biomedical question answering, particularly to such an area as clinical decision support and medical education tools where a short and precise answer is quite necessary.

Xia et al. (2022) [10] created LingYi, a multi-modal conversational medical question answering system that conducted textual and visual processing by using a multi-modal knowledge graph. In contrast to the traditional QA systems, which can only focus on the text retrieval, LingYi uses both structured and unstructured medical data of various origins such as textual clinical notes, and diagnostic images like X-rays. The architecture of the system deals with three primary components namely, entity recognition, multimodal alignment and a dialogue manager of a dynamic question answering system. It leverages natural language processing models in visual semantic segmentation tools to develop thorough medical knowledge graphs to support more comprehensive capabilities in inference. LingYi is especially beneficial when used in medical settings where visual evidence has been shown to be an important part of diagnosis and planning of further treatment. As an example, the user may pose a question regarding a chest X-ray image and the system may consult on features of images and integrate specific text information in a bid to create an answer. The system can have multi-turn conversation which helps to resemble the natural conversational doctor-patient dialogue. Yet, multimodal data combination requires extensive computing resources and custom preprocessing pipelines and might introduce some limitations that could constrain the real-time scale and implementation. In addition to that, training of such systems implies the use of annotated datasets with aligned visual-textual data, which are limited in the medical sector. Despite them, LingYi is also an essential step towards more human-like and overall medical dialogue systems.

He et al. (2020) [11] came up with a new method to aid the use of BERT in the medical domain, filling it with specific knowledge, which would be represented by the structure, which created the model categorised as Knowledge-Infused BERT (KI-BERT). The main reason why such approach is applied lies in the fact that in many cases of general-purpose pre-trained models such as BERT have little to no profound knowledge of medical terms and relations. To overcome this, the authors have embedded the knowledge concerning the disease into the embeddings of the model directly using existing biomedical ontologies that include disease information like the Unified Medical Language System (UMLS). This was brought about by a pretraining approach that harmonises the text four content of questions with systematic medical knowledge enabling the model to understand and infer medical concept like medical entities, medical signs, and medical interventions significantly better. Various tasks were conducted on the resulting model such as medical question answering, disease name recognition, and health-related inference. The model consistently performed better than vanilla BERT. Impressively, KI-BERT performed very well when applied to interpret contextually similar terms of diseases and in the identification of paraphrase or misspelled medical questions. Among the most important contributions of this work is its ability to show how domain knowledge can be reproduced systematically into language models to greatly enhance performance in specialized areas of interest. Nonetheless, its limitation is that it only focuses on disease-related tasks which may not cross over that well to other subtopics of medicine like pharmacology or radiology. Also, curated ontologies play a very important role in the system, thereby potential development opportunities are to miss the emerging terms or patient language.

Yagnik et al. (2024) [12] have introduced MedLM, a thorough study of using large language models (LLMs) in medical question answering, with specific attention paid to the process of instruction tuning and prompt engineering to enhance the correctness of answers and their relation to context. They assessed a wide range of open-source and proprietary LLMs like GPT-3.5, LLaMA-2 and PaLM on many medical QA datasets. The authors focused on the necessity of aligning model responses provided to the domain-specific expectations that were considering in their work based on the individual tuning of instruction by multi-turn medical dialogues and task-specific templates. Evaluation was done based on the BLEU, ROUGE, and human expert metrics, which showed that instruction-tuned models were much more accurate in the single-answer and conversational question-answer tasks. Moreover, MedLM has a benchmarking framework, which allows the researchers to evaluate model behaviour regarding correctness, coherence, and safety as the most critical issues in healthcare applications. One of the most interesting points in this research is the commonness of hallucination even in the uppermost LLMs that dearly raise the question of their use in practical medical care without a professional supervisor. The authors suggest hybrid systems in which LLMs are combined with verified medical bodies of knowledge or mechanisms of human-in-the-loop. In general, MedLM offers a basis inspiration of how the latest variety of LLMs can be carefully tailored to healthcare, however, it also highlights several ethical and technical challenges which are still there, including explainability, factual accuracy and auditing the model.

Qiu et al. (2025) [13] successfully illustrated that the aim of this study is to enhance the medical visual question answering system by proposing interpretable frameworks to incorporate chain-of-evidence reasoning. The data used comes from the PMC-OA corpus, which consists of 1.6 million medical image-title pairs of different modalities and diseases from PubMed centers. In the data preprocessing stage, the medical images will be resized to an image size of 224x224 by interpolation (bicubic interpolation), the text captions and descriptions will be tokenized using Flan-Alpaca's SentencePiece tokenizer, and in the case of multiple-image instances, the first image should be selected each time to process them consistently. The two knowledge sources used to build the knowledge graph are the PMC-OA Evidence Context which provides paired medical images and text and Wikipedia Knowledge which links to validated medical texts on Wikipedia, and the model will learn semantic matches between the visual features and the corresponding text descriptions. The model was trained in three steps such as pre-training of the medical visual language on the PMC-OA dataset, an inference step using GPT-3.5 to generate chains of evidence to generate auto-labeling and fine-tuning of the

model using a dual-encoder architecture consisting of the ViT-B/16 vision converter and the Flan-Alpaca text encoder. Accuracy and BLEU were used to measure the performance of the system. In the open-ended task, the system had an accuracy of 43.4%, and in the multiple-choice task, the system had an accuracy of 87.4%. The advantage of the system is that verifiable medical reasoning can be generated because the responses are based on two sources of medical knowledge and the links between the evidence can be traced. The disadvantages of the system are that it requires GPT-3.5 to generate the chain of evidence during training, which can lead to factual errors, and that the model is relatively small (0.4B parameters) compared to other large language models.

Zafar et al. (2024) [14] researched that this work is done to develop a Knowledge-Infused Medical Abstractive Generator (KI-MAG) model to improve abstractive question-answering performance in the medical domain by addressing data sparsity and hallucination issues. The data was collected through BioASQ Task 10b Phase B, where 4,232 biomedical Q&A pairs could be found in the dataset, and the data was synthesized using the T5 question generation model to generate a new set of 30,172 QA pairs named BioASQ-SYN. In the data preprocessing phase, text cleaning and tokenization was performed through the BART-Large model, and medical entity extraction was performed through the Scispacy library for medical entity extraction, and a pipeline for generating synthetic data via the Google Translate API, which allows context selection and paraphrasing based on medical entity density. Knowledge characterisation was achieved by constructing a medical knowledge graph based on the Unified Medical Language System (UMLS), which contains about 2 million medical concepts and 12 million interactions, and employing KG filtering or pruning strategies based on cosine similarity to obtain the top 60 most relevant triples for each question. It was trained using BART-Large as a backbone with a KG-enhanced encoder with Q2K (Question-to-KG) about Q2K (Question-to-KG) and K2Q (KG-to-Question) tons and was trained using the Adam optimizer for 50 epochs of synthetic dataset and 15 epochs of BioASQ data using the Adam optimizer. BLEU score values (15% increase), ROUGE-L, METEOR, medical entity F1 scores, as well as fluency and adequacy metrics for human assessments and medical entity correlations were used to measure system performance. The advantage of this system is that it can combine textual and structured knowledge well through a novel attentional mechanism, which greatly reduces illusions and improves the accuracy of answers, especially in the area of quality assurance in the medical field. The disadvantages of such systems are that they depend on the quality of the UMLS knowledge graph and that the system generates lengthy answers to fact-based questions due to the synthetic type of training data.

Liu et al. (2024) [15] proposed that this research is done to develop a Parameter-Efficient Transfer Learning framework called VQA-Adapter for medical visual question answering (Med-VQA) to overcome computational costs and improve performance compared to full fine-tuning approaches. Their data comes from two well-known Med-VQA datasets: the VQA-RAD dataset contains 315 medical images and 3,515 question answers, and the SLAKE-EN dataset contains 642 medical images and 7,032 question answers. In the data preprocessing stage, CLIP-ViT-Base-32 was used to freeze the visual coder, GloVe was used to initialise the word embeddings, and LSTM was used to extract the semantic information of the questions. The representation of knowledge is implemented by a lightweight visual adapter module that includes a layer for down sampling and up sampling the frozen CLIP, and multimodal feature fusion between the visual and textual representations is done using a BAN (Bilinear Attention Network). The model is built on frozen CLIP-ViT, and the only trainable part is the proposed adapter module, which requires only 2.38% parameter modification instead of 100% fine-tuning, while multilevel label smoothing with varying ratios is used in the training phase to avoid overfitting. The performance of the system was evaluated by calculating the accuracy results for both open-ended and closed-ended problems, which were 75.8% for VQA-RAD and 81.0% for SLAKE-EN, which is an improvement of 3.8% and 0.9%, respectively, compared to the state-of-the-art system. The advantage of this system is that it is parametrically efficient, but the performance is kept at a high level because it occupies much less computational resources compared to a fully fine-tuned system. The disadvantage of the system is that it relies on advantages in terms of frozen CLIP, which in some cases is unlikely to understand medical information in each domain, and therefore the model itself gives inaccurate answers to complex medical problems involving sophisticated reasoning methods.

The study by Zhou et al. (2025) [16] demonstrated that it created a TCM Lung Cancer Intelligent Question and Answer Model (TCMLCM) by combining a large-scale language model with a structured knowledge graph through an approach called KG2TRAG. Data were obtained from three open TCM data repositories like Tianchi TCM, Huangdi, and Shennong TCM datasets, and a TCM Lung Cancer Knowledge Graph (TCMLCM) was developed on the open resources, which contained 11,723 independent nodes and 15,912 edges to represent the relationships among diseases, evidence, formulas, and herbs. In the data preprocessing step, repetitive entity descriptions are eliminated by entity alignment technique based on semantic and textual similarities, and the data in the knowledge graph are built in the form of vectors and stored in a vector database using the BGE model. KG2TRAG is used to represent the knowledge in the form of a knowledge representation in which the knowledge graph triples are natural-language with the help of the ChatGPT-assisted linearisation representation to improve the retrieval efficiency. The model was trained with ChatGLM2-6B as the starting model with 8-bit quantization bit-width, fine-tuned using LoRA technique, and trained on 50672 fine-tuned data pairs for 3000 steps at a learning rate of 2E-3. By measuring BLEU (32.15%), ROUGE-L (59.08%), and accuracy (79.68%), TCM tumour professionals demonstrated the quality of answers, professionalism, and usefulness of the system. The most important feature of the system is its ability to successfully combine structured medical knowledge with contextual understanding of words and fine-tune them in a parameter-efficient manner, thus requiring only a small amount of computational resources. The shortcomings of the system are that the decision-making

process remains a black box for the user, and that rare prescriptions or specific therapies will be reflected in the patterns of ancient Chinese medicine due to the model's tendency to favour high-frequency patterns.

Table 1. Summary of Related Works.

Author	Objective	Model Used	Strength	Weakness
Deepa et al. (2024) [1]	Improve medical QA recommendation performance.	GPT-3.5 Turbo-0125	The system generates strong recommendations.	Fine-tuning is done on basic models.
Alruwaithi & Alhumoud (2025) [2]	Enhance existing Arabic healthcare question answering system.	SBERT + Classifier models	It achieves high accuracy compared to other models.	Lacks testing of other NLP applications.
Sukhwal et al. (2024) [3]	Design question-answering system for disease-related queries.	Large Google T5 LLM	Combines KG and LLM effectively to provide high quality answers.	It cannot update with new medical knowledge.
Q. Peng et al. (2024) [4]	Develop medical question-answering system that is based on multi-source data.	Medical Multi-Agent (MMA)	Produce structured decision-making output.	Lack of knowledge causes difficulty in solving complex task.
Alkhurayyif (2025) [5]	Build an Arabic OQAS to improve health information accessibility.	GCN + GPT 2.0	High accuracy for complex Arabic medical questions.	Issues with Arabic script, code-switching, and inconsistent spellings.
Das & Nirmala (2022) [6]	Improve healthcare question answering accuracy using domain-specific deep learning.	BioBERT	Perform strong understanding of medical terms.	Rely on SQUAD dataset, which is not medical specific.
Vinod et al. (2021) [7]	Provide multilingual medical question answering for rural healthcare access.	BioBERT + LABSE + GPT-2 + DistillBERT	Multilingual and can be used offline.	HER summarisation part is still under active development.
Jeong et al. (2023) [8]	Automate healthcare question answering for online platforms.	BERT + GPT-2	Reduce repetitive manual work for doctors.	Has higher perplexity and loss, which may be less accurate.
Nguyen et al. (2022) [9]	To develop a two-stage system of QA to achieve higher precision in response to medical questions with sentence-level retrieval and deep learning.	SPBERT + fine-tuned BERT	Obtains a high level of precision in retrieving semantic answers to medical texts.	Context is restricted to sentence level; search fails to find an answer in multiple sentences.
Xia et al. (2022) [10]	To create a conversational medical QA system to integrate visual and textual data through a multi-modal knowledge graph.	Multi-modal knowledge graph + NLP modules	Interactive multi-turn dialogues and answers to both image and text are supported.	Requires high computational resources and multimodal annotated data.
He et al. (2020) [11]	To improve the BERT model to make it specifically suitable to medical tasks by infusing biomedical ontologies as disease knowledge.	Knowledge-Infused BERT (KI-BERT)	Enhances the knowledge between entities and queries of disease.	Restricted to the disease domain; the state of medical ontologies relies considerably on the frozen ontologies.
Yagnik et al. (2024) [12]	To investigate the instruction-tuned LLMs to achieve more consistent and secure medical QA systems.	GPT-3.5, LLaMA-2 + Instruction Tuning	Contextual precision and less hallucination are brought about by custom prompts.	Without any human supervision, models can continue hallucinating or delivering information that cannot be proved.
Qiu et al. (2025) [13]	Improve MedVQA through explainable chain-of-evidence reasoning	Med-CoE (Flan-Alpaca + ViT-B/16 + GPT-3.5)	Verifiable reasoning with dual knowledge sources; traceable evidence links.	Reliance on GPT-3.5 may introduce errors; small model size (0.4B).
Zafar et al. (2024) [14]	To develop a knowledge-infused abstractive QA system for medical domain that addresses data sparsity and hallucination issues	KI-MAG (BART-Large with KG-augmented encoder using Q2K and K2Q attention mechanisms)	Effectively combines textual and structured knowledge, reduces hallucination, achieves 15% improvement in BLEU scores.	Depends on UMLS knowledge graph quality, generates longer answers for factoid questions.
Liu et al. (2024) [15]	To develop parameter-efficient transfer learning for Med-VQA to reduce computational costs while maintaining high performance	VQA-Adapter (frozen CLIP-ViT with trainable adapter module + BAN for feature fusion + multi-stage label smoothing)	Achieves superior performance with only 2.38% parameter updates, significantly reduces computational costs.	Depends on frozen CLIP features quality, may struggle with complex medical reasoning tasks.
Zhou et al. (2025) [16]	To develop an intelligent QA model for TCM lung cancer by integrating LLMs with KGs using KG2TRAG method	TCMLCM (ChatGLM2-6B with LoRA fine-tuning + KG2TRAG method for knowledge graph integration)	Effectively integrates structured knowledge with language understanding, achieves high performance with less computational cost.	Decision-making process lacks transparency, may overlook rare prescriptions in ancient TCM texts.

3 Data Collection

The data collection process starts with identifying relevant and suitable online sources such as public health websites that have dengue-related information. The sources are chosen to ensure that it is appropriate and accurate for our system. Information from the online websites is gathered and extracted manually. This is done to ensure that the quality of data is reliable as it only ensure the relevant information is extracted. The non-relevant content like the navigation menu, advertisements, comments and unrelated site sections are filtered out. The extracted information is saved in a CSV file named "Dengue_QA.csv" with four columns such as Category, Question (Malay), Answer (Malay) and Source URL. The category refers to the type of information extracted such as dengue descriptions, symptoms, treatment methods and prevention measures. As for the row, it has 149 rows which represents a question-answer pair in Malay language. This file is then reviewed to make sure that it only contains dengue-related information and does not have any irrelevant information. The data is also ensured to be accurate, relevant and detailed based on its entries in the categories.

4 Data Pre-processing

The data pre-processing is the crucial step in NLP projects like our intelligent question answering system for dengue-related information. The main objective of this stage is to process the raw text data into a clean, structured, and suitable format. Therefore, the quality of data could be improved for the following knowledge representation and model training phases. This stage is important especially when dealing with multilingual data due to inconsistences and noise. The initial dataset consists of paired question-answer entries in both Malay and English. Each pair of question and answer are associated with a category and its source URL in a CSV file named "Dengue_QA_multilingual.csv". The data pre-processing begins with read and manage the dataset using the pandas library.

4.1 Duplicate and Missing Data Removal

To ensure that there is no repeated or redundant data in the dataset, this work used the "drop_duplicates" function in pandas. This function scans the entire dataset automatically, which finds the rows with the same column values, keeps only one unique row, and remove all the duplicates one. This is because the duplicates may affect the learning process of model and evaluation performance of the system. In addition, missing data can reduce the response quality of the question answering system. This study checked the dataset and removed the records that were missing questions or answering in Malay or English. Thus, each data entry contained a complete bilingual question and answer pair, enhancing the reliability and performance of this dengue QA system.

4.2 Text Cleaning

Since the texts may contain some incorrect word forms and variations, the text cleaning stage is required to remove irrelevant content. This process is applied only to the question columns as these columns contain the raw textual input that the system or model needs to understand and process. To clean up these texts, this work used regular expressions to remove all the HTML tags, which contain the HTML elements like <b>,
, <p>, and so on. This ensure that only plain text is retained in the Question (Malay) and Question (English) columns. After that, special characters and punctuation such as @, #, $, and *, which are meaningless for answering questions, are removed. This step used also the regular expressions to remove those characters but remained necessary alphanumeric characters. The whitespace characters are normalised, and the consecutive spaces, tabs, or line breaks are replaced with a single space. The leading and trailing spaces are removed to ensure that the text formatting is consistent for the subsequent processing.

4.3 Answer Formatting

The answer format addresses the need to present multi-part answers, which are common with health questions that often include many steps or recommendations. To facilitate readability and consistency of system, each subsequent line of an answer is formatted using a tilde character, "~". Regardless of the format used for the first answer, this method ensures uniform representation, simplifies downstream processing or display of multi-line answers, and individuates points. Whenever responses are shown to users, this type of writing is extremely helpful since it simplifies challenging responses by showing bullet-point clarity.

4.4 Lowercasing and Tokenisation

Every Malay and English question was brought into lowercase to avoid discrepancies caused by different capitalisation techniques. For example, "Denggi" and "denggi" is addressed by lowercasing. This process minimises the vocabulary size and facilitates further analysis of the text by making words that differ only in capitalisation the same token. In most NLP tasks, tokenisation refers to dividing text into individual basic units, which called tokens. For Malay text, the Malaya library tokeniser, which builds specifically to handle the complex language structure of Malay, is used. The English text is processed using the "word_tokenize" function from the Natural Language Toolkit (NLTK) library. By converting the original sentence into a list of tokens, tokenisation enables to perform the following linguistic operations such as stopword removal, stemming and lemmatisation.

4.5 Stopword Removal

Stopwords are common words such as "the", "is", "yang", and "adalah" that appear frequently in the text but have limited semantic content and cannot identify important information. Stopword elimination helps focus on important content words, reduce computational costs, and improve model performance, especially in keyword-based retrieval and matching tasks. For Malay, this work used the "get_stopwords" function of the Malaya library to handle stop words, which returns a complete language-specific list. For English, the stop word list pre-computed by NLTK. After stop word removal, only the most precious terms are left, so that subsequent processing has precious materials to use.

4.6 Stemming and Lemmatisation

Natural languages tend to use different inflected forms of words like "diagnose," "diagnosing," and "diagnosed." Lemmatization and stemming are the language processing process of reducing words to their base or root form. Since Malay is a highly inflected language, its stemming is done through the Malay Sastrawi stemmer. This process enables multiple grammatical forms of a word to be recognised as having the same meaning. English lemmatisation is done through the NLTK WordNet lemmatiser, which reduces words to their dictionary base form, but fits the context of the word. Normalisation is necessary to correctly match user queries to knowledge base records despite grammatical differences.

5 Knowledge Representation

To create a knowledge graph, the process starts by importing a CSV file named “Dengue_QA_multilingual_cleaned.csv” which contains question-answer pairs in both Malay and English language. This CSV file is the read into pandas DataFrame which is then converted to a list of Python dictionaries where each of it represents a question-answer pair. The list is saved in JSON format named “Dengue_QA_multilingual.json” to ensure that the data can be easily used in web application. This is because in JSON, each question-answer pair is represented as object that contains category, question, answer and source which makes it easy to process.

Next, RDFLib library is used to construct the knowledge graph based on the CSV data. This is created to ensure that the knowledge graph can make relationships between categories, questions and answers easier to understand by machine and queryable. The namespace, EX = Namespace("http://example.org/dengue#") is defined to create a unique URIs for each question-answer pair. A unique node with its unique id is created to represent each pair in the DataFrame. The nodes represent the question and answer in both Malay and English language along with its category and the source URL. These attributes are then added as RDF triples such as subject, predicate and object which then form a complete RDF graph that is saved in Turtle format named “Dengue_QA_Multilingual_KG.ttl”.

There are two libraries which are NetworkX and Matplotlib used to visualise the knowledge graph. The nodes are created for Malay and English questions along with its answers and category. Then, edges are added from questions to answers and questions to category with the label hasAnswer and hasCategory respectively. The graph is then shown using a force-directed layout known as spring_layout where the nodes and edges are drawn with labels and has title above.

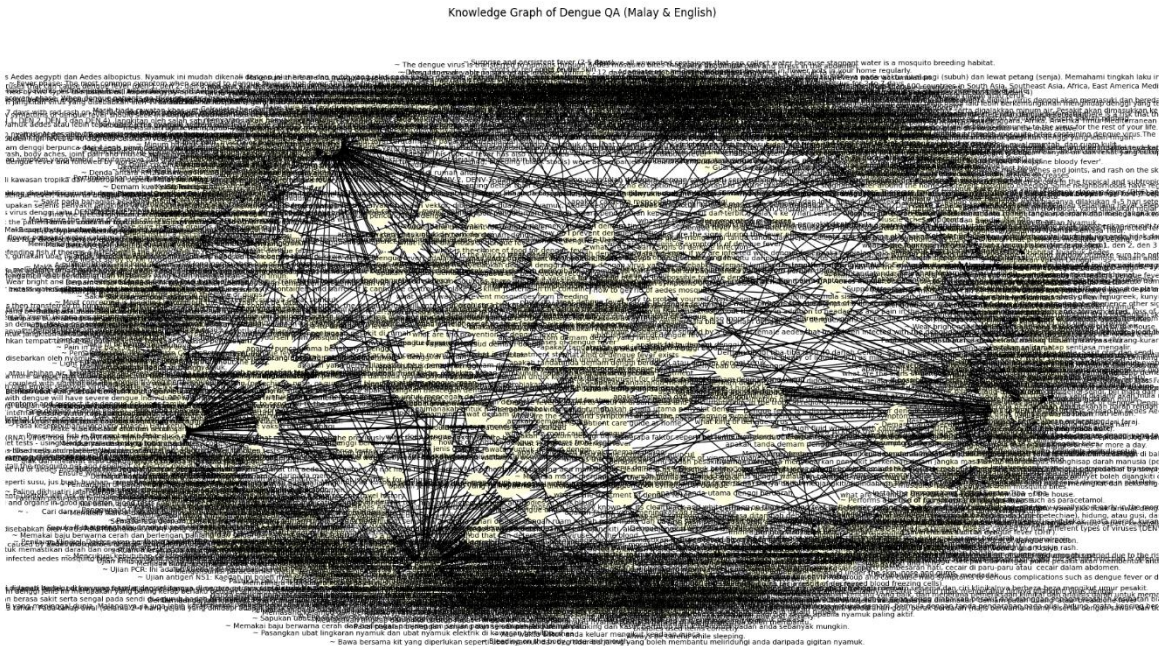


Figure 1. Knowledge Graph of QA pairs

In the graph above, it can be seen that it is a dense graph that looks like hairball due to many nodes and edges overlapping each other. The nodes represent every question and answer in Malay and English language along with its categories. The edges such as arrows or lines show the relationships between them. To conclude, the graph is just a visualisation to show how the question-answer data is structured and interconnected. As the purpose of knowledge graph is to enhance machine-readable triples for semantics queries and not for illustration understanding.

Therefore, a smaller subset of knowledge graph is also visualised to improve readability. It has the same process as the above knowledge graph but it is limited to few nodes and edges which makes it clearer and easier to interpret. This is because it shows a one-to-one relationship between the pair and highlight the accurate linkage through the edges.

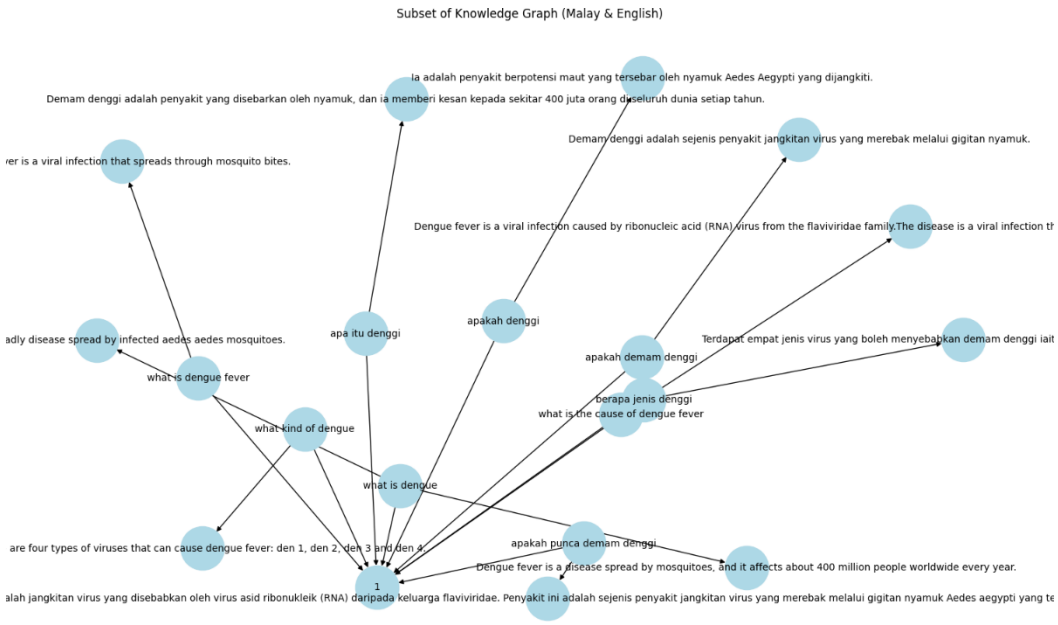


Figure 2. Subset of Knowledge Graph of QA pairs

Finally, the knowledge graph in Turtle format is loaded using RDFLib. The knowledge graph is used to run SPARQL query where it can retrieve the number of question-answer pairs based on each category. The output is shown where each category has the relevant number of pairs such as 47 pairs for Category 1, 37 pairs for Category 2, 35 pairs for Category 3 and 30 pairs for Category 4. This sums up to 149 question-answer pairs which was collected earlier. This shows that the data can be queried in flexible ways when it is in the form of knowledge graph.

To conclude, the knowledge graph is loaded together with the CSV and JSON data. The system can retrieve answers from the pairs directly when user asks question. It can also query the graph for related entities even if the exact wording is not found. This is because it infers new answers based on the relationship through the use of this knowledge graph. This knowledge graph enhances the coverage of answer and its ability to handle user queries.

6 Model Training

In this project, the model training phase focused on constructing a reliable question answering system by generating sentence embeddings and integrating them into a semantic similarity matching pipeline. Two different multilingual pre-trained sentence transformer models were selected and evaluated to determine which model provided the most accurate and contextually appropriate answers to user queries. Both models were incorporated into the system, and their performance was compared using the same dataset. The model with higher accuracy and retrieval performance was subsequently selected and implemented as the final QA system.

6.1 Model Selection and Architecture

The core approach for this QA system leverages sentence embeddings to enable semantic retrieval of answers based on user queries. Two prominent multilingual sentence transformer models such as paraphrase-multilingual-MiniLM-L12-v2 and paraphrase-multilingual-mpnet-base-v2 were selected for training and evaluation. Both models are specifically designed to handle multiple languages including Malay and English, making them ideal for the bilingual nature of this dengue information system.

The MiniLM-L12-v2 model, with its 384-dimensional embeddings, offers a lightweight yet effective solution for semantic similarity tasks. In contrast, the mpnet-base-v2 model produces 768-dimensional embeddings, potentially capturing more nuanced semantic relationships at the cost of increased computational requirements. Both models utilise

the transformer architecture and have been pre-trained on extensive multilingual datasets, enabling them to understand and encode semantic meaning across different languages.

6.2 Training Methodology

The training process followed a rigorous methodology to ensure reliable model performance evaluation. The dataset of 149 QA pairs was split into training and testing sets using a 70-30 ratio with stratified sampling to maintain category distribution. This resulted in 104 training samples and 45 testing samples, with balanced representation across all four dengue information categories.

The system construction process involved generating sentence embeddings for all questions in both Malay and English using the selected pre-trained sentence transformer models. These embeddings were computed with a batch size of 32 and stored as numpy arrays for efficient similarity calculations. The system employs cosine similarity as the primary metric for matching user queries to the most relevant questions in the knowledge base.

6.3 Retrieval Strategy Enhancement

To improve the accuracy of answer retrieval, a Top-3 retrieval strategy was implemented. Instead of relying solely on the single most similar result, the system retrieves the top three candidates based on cosine similarity scores. The selection process then prioritises answers whose categories match the expected category, falling back to the highest similarity score if no category match is found. This significantly increases the likelihood of returning contextually appropriate answers, as reflected by the high category classification accuracy achieved during evaluation.

6.4 Model Evaluation and Performance

Both models were thoroughly tested on unseen test sets to examine their performance in real-life situations. Comparative metrics include exact match accuracy, average similarity, retrieval success (similarity greater than 0.5), category accuracy, and standard categorisation performance metrics like precision, recall and F1 score. The models were cross-tested using Malay and English questions to ensure a delicate balance of their multilingual performance.

In the paraphrase-multilingual-MiniLM-L12-v2 model, the success rate of category classification is 94.44%, and the average similarity score is 0.9202. The success rate of the retrieval results is 94.44%, which reflects a strong understanding of the semantics. In the paraphrase-multilingual-mpnet-base-v2 model, the categorisation accuracy is 95.56%, the average similarity score is 0.9083, and the retrieval success rate is 97.78%.

6.5 Model Comparison and Selection

A detailed comparison of both models revealed that while both demonstrated excellent performance, the mpnet-base-v2 model showed marginal improvements in key metrics. The comparison table below summarises the performance differences.

Table 2. Performance Comparison between MiniLM-L12-v2 and mpnet-base-v2 Models.

Metric	MiniLM-L12-v2	mpnet-base-v2
Model Name	paraphrase-multilingual-MiniLM-L12-v2	paraphrase-multilingual-mpnet-base-v2
Embedding Dimension	384	768
Category Accuracy	94.44%	95.56%
Retrieval Success Rate	94.44%	97.78%
Average Similarity	0.9202	0.9083
Precision (weighted)	0.9445	0.9567
Recall (weighted)	0.9444	0.9556
F1-Score (weighted)	0.9443	0.9556
Model File Size	0.37 MB	0.67 MB

Despite the larger file size, the mpnet-base-v2 model was selected for deployment due to its superior performance metrics, particularly the higher retrieval success rate and category classification accuracy. The model's ability to maintain high performance across both languages while providing more reliable retrieval results justified the additional computational overhead.

6.6 Training Outcomes

The model training phase successfully produced a robust multilingual QA system capable of understanding and retrieving relevant dengue information in both Malay and English. The implementation of the Top-3 retrieval strategy, combined with the superior performance of the mpnet-base-v2 model, ensures that users receive accurate and contextually appropriate answers to their queries. The high category classification accuracy demonstrates that the system can effectively distinguish between different types of dengue-related information, from general descriptions to specific prevention methods.

The final trained model, saved as a serialised pickle file, includes the pre-computed embeddings for all training questions, enabling rapid similarity calculations during runtime. This approach optimises the system's response time while maintaining the high accuracy standards required for a healthcare information system.

7 Deployment

In this work, Gradio was adopted as the user interface layer, while Hugging Face Spaces was chosen as the hosting solution. Gardio is an open-source Python library that can simplify building interactive web interface, which is well-suited for the question answering application. Besides, Hugging Face Spaces is a platform that enables to easily create and deploy the applications. This platform is very useful as it does not need to perform any software installation or manual configuration. By using this integration, the QA system can be accessed conveniently from any web browser.

During the development process, there are two multilingual Transformer models, which are paraphrase-multilingual-MiniLM-L12-v2 and paraphrase-multilingual-mpnet-base-v2, have been trained and evaluated. After a comparative analysis, the mpnet-base-v2 model was selected for deployment. The overall performance of this model was more stable, proved by the higher category accuracy and retrieval success rate. Moreover, the semantic embeddings and related question-answering data generated by mpnet-based-v2 model are serialised and stored in a single file to load and query quickly at runtime.

The deployment process begins with the preparation of all applications files, including the main Python script (app.py), model file (dengue_qa_model (mpnet-base-v2).pkl), and requirement statements (requirements.txt). After that. all these files are uploaded to a new created Hugging Face Space configured with the Gradio SDK. Once these files are uploaded, the Hugging Face Space starts the environment build and installs all specified Python packages automatically. The application is launched and assigned a public URL when the installation is successful. This URL serves as a primary access point, which users can access this URL from any device equipped with a web browser. Through this web interface, users can submit the dengue-related questions either in Malay or English and receive accurate and consistent responses. For this project, the intelligent QA system for dengue related information with multilingual can be accessed at https://huggingface.co/spaces/ly-17/Lab1B-G5.

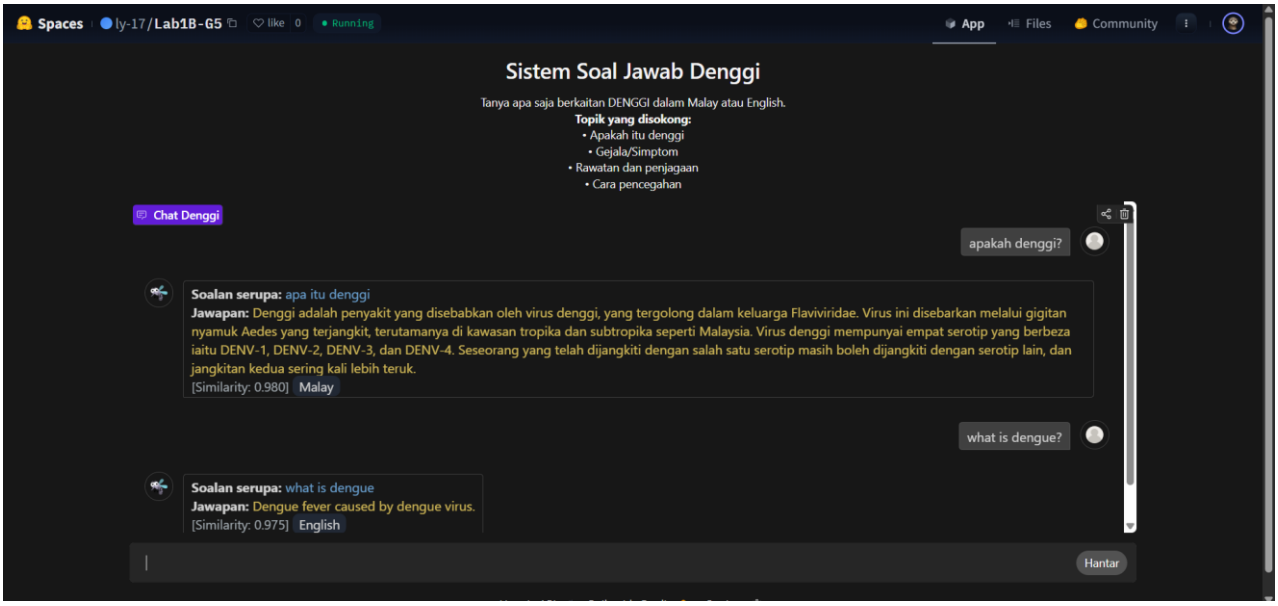


Figure 3. Sample Screenshot of QA system

8 Advanced Features

8.1 Knowledge Graph Integration

This question answering system integrate knowledge graph as an advanced feature. The knowledge graph is built using RDF and stored in Turtle format. It represents the pair of questions and answers along with their semantic relationships which are multilingual questions and answers, categories and sources. This improves the advance querying and visualisation of answer retrieval accuracy. This integration is advanced because it has semantic understanding where it does not only rely on simple text retrieval as it can connect the entities or categories and their relationships. It also has reasoning ability where it can infer answers based on synonyms, paraphrasing and complex queries if the question does not exactly match the dataset.

8.2 Multi-lingual Question Answering

The multi-lingual question answering is implemented with pandas library that handles the CSV data. It employed an automatic translation library known as googletrans which can perform Malay-English translation. The original dataset that is collected in Malay language is loaded. Then, the process of translating the question-answer pair from Malay to English language starts by iterating through each row. The process involves automated translator which is either Google Translate API or pre-trained translation model. The translated data is then stored in new columns known as Question (English) and Answer (English). The data is checked to ensure that the meaning are accurately represented. The DataFrame is saved as a new CSV file named "Dengue_QA_multilingual.csv" which has six columns in total. The columns are Category, Question (Malay), Answer (Malay), Source URL, Question (English) and Answer (English). This ensures that the system can handle questions in both Malay and English language which is also known as the term "rojak". The flow of the process is shown below.

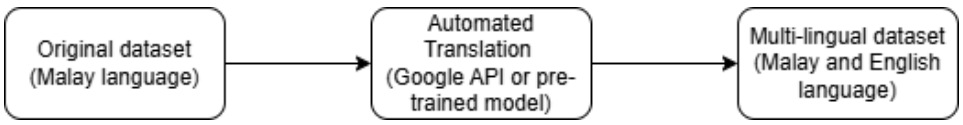


Figure 4. Multilingual Process

9 Conclusion

In summary, this project has been able to come up with an intelligent question answering (QA) system capable of getting quality and affordable health information related to dengue. The system can be used to improve health stills in both English and Malay languages because of the need of dependable health communication resources. The manually curated and multilingual dataset along with the usage of a semantic knowledge graph enables the system to provide an accurate answer to any user query and understand the context better.

The combination of RDF-based knowledge graphs allows structuring pairs of questions and answers into machine-readable and structured triples, so the system can run advanced SPARQL queries to retrieve information in a sufficiently flexible way. NetworkX along with Matplotlib visualisation assists in the interpretation of the intricate graph structure and displays the relationship among categories, questions and answers. Moreover, the system supports multilingual capabilities using translation tools. This makes it able to be communicated to larger groups of people and non-English speakers which is critical aspect in the multilingual environment of Malaysia.

The results of this project not only prove the possibility to address public health awareness using intelligent QA systems but also show the importance of the combination of the structured knowledge representation approach and the NLP techniques. This system can be the base of future improvements including real-time integration, voice-based interaction, or even integration with the databases of the public health sector and provide the updated information automatically. This system thus plays a significant role in making the health information user-friendly, accessible, and interpretable to a greater extent alongside the growth of the dependency on tools saturated with artificial intelligence.

Finally, the study of the intelligent QA system of dengue-related information proves the potential usage of artificial intelligence in the real world around us, having social significance. It would open the way to establishing stronger and domain-specific QA systems that might benefit other spheres of the public and services of public health, education, or government to ensure more digital inclusivity and knowledgeable decision-making.

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